Collective Risk Modelling

The individual Risk model

- $S = Y_1 + \dots + Y_n = \sum_{i=1}^{n} Y_i$

M2

Convolutions

- $F_{X+Y} = F_X * F_Y$
- $F_{X+Y+Z} = F_X * F_{Y+Z}$

discrete

- df: $F_{X+Y}(s) = \sum F_Y(s-x) f_X(x)$
- pmf: $f_{X+Y}(s) = \sum f_Y(s-x) f_X(x)$

continuous

- df: $F_{X+Y}(s) = \int F_Y(s-x) f_X(x)\,dx$
- pdf: $f_{X+Y}(s) = \int f_Y(s-x) f_X(x)\,dx$

MGF and PGF

- If independent: $M_S(t) = M_{Y_1}(t) \cdot M_{Y_2}(t) \cdot \dots \cdot M_{Y_n}(t)$
- Same as PGF

$S = \sum_{i=1}^{N} Y_i$

- Y_i and N mutually independent
- $E[S] = E[N] \cdot E[Y]$
- $Var(S) = E[N] E[Y^2] + E[Y]^2 (Var(N) - E[N])$
- $M_S(t) = M_N(\log(M_Y(t)))$

Compound Binomial $Var < E[S]$

- $S \sim CompBinom(v, p, G)$
  - $\to v, p$ are parameters
  - $\to G$ is claim size distribution
- $S = \sum_{j=1}^{n} S_j \sim CompBinom(\sum v_j, p, G)$

Compound Poisson $Var > E[S]$

- $S \sim CompPois(\lambda v, G)$
- Properties:
  - Aggregate property $\to$ bottom-up
  - Disjoint property $\to$ top-down

Mixed Poisson

- Assume $\Delta \sim H$, $H(0) = 0$, $E[\Delta] = 1$, $Var(\Delta) > 0$
- for a given Δ, $N \sim Pois(\Delta v)$ for $v > 0$
- $P(N=n) = \int_0^\infty e^{-\lambda v x} \frac{(\lambda v x)^n}{n!}\, dH(x)$
- $E[N] = \lambda v$
- $Var(N) > \lambda v$
- $M_N(t) = M_\Delta(v(e^t - 1))$

Compound Negative Binomial

- $S \sim CompNB(\lambda v, \gamma, G)$

$Var(N) > E[N]$

M- dispersion

Properties of Poisson frequencies

Aggregation Property ↑

- Assume $S_1, \dots, S_n$ are independent with $S_i \sim CompPoi(\lambda_i, v_i, G_i)$, $i = 1, \dots, n$

$S = \sum_{i=1}^{n} S_i \sim CompPoi(\lambda v, G)$

$v = \sum_{i=1}^{n} v_i$, $\lambda = \sum_{i=1}^{n} \frac{v_i}{v} \lambda_i$, $G = \sum_{i=1}^{n} \frac{\lambda_i v_i}{\lambda v} G_i$

Disjoint Decomposition ↓

- Assume S is "doubly partitioned"
- $\to S_{A} = \sum_{i=1}^{N} Y_i \mathbb{1}_{\{(Y_i, I_i) \in A_k\}} \sim CompPoi(\lambda_k v, G_k)$
- $\to \lambda_k v = \lambda v p^{(k)} > 0$, $G_k(y) = P(Y_1 \le y \mid (Y_1, I_1) \in A_k)$

Sparse Vector algorithm

- If $S \sim CompPois(\lambda, g(y_i) = \pi_i)$, $i = 1, \dots, m$
  then $S = y_1 N_1 + \dots + y_m N_m$
  $N_1 \perp \dots \perp N_m$, $N_i \sim Poi(\lambda_i = \lambda \pi_i)$
  $P_i(\theta) = P(iN_i = \theta) = P(N_i = \frac{\theta}{i})$

Large Claim Separation

- Let M be claim threshold $S_{lc} = \sum_{i=1}^{N} Y_i \mathbb{1}_{\{Y_i \le M\}}$
- $S_{lc} = \sum_{i=1}^{N} Y_i \mathbb{1}_{\{Y_i > M\}}$
- $S = S_{sc} + S_{lc}$

$S_{sc} \sim CompPois(\lambda_{sc} v = \lambda G(M) v, G_{sc}(y) = P(Y_1 \le y \mid Y_1 \le M))$

$S_{lc} \sim CompPois(\lambda_{lc} v = \lambda(1 - G(M)) v, G_{lc}(y) = P(Y_1 \le y \mid Y_1 > M))$

Copula

Pearson's correlation coefficient

- $\rho(Z_i, Z_j) = \rho_{ij} = \frac{Cov(Z_i, Z_j)}{\sqrt{Var(Z_i) Var(Z_j)}}$
- $\rho \in (-1, 1)$ and measures the degree of linear relationship between Z_i and Z_j

Kendall's tau

$\tau(Z_i, Z_j) = \tau_{ij} = 4E[F(Z_i, Z_j)] - 1$

$\tau_{ij} = P[(Z_i - Z_i')(Z_j - Z_j') > 0] - P[(Z_i - Z_i')(Z_j - Z_j') < 0]$

Spearman's rho

- $r(Z_i, Z_j) = r_{ij} = \rho(F_i(Z_i), F_j(Z_j))$
- Where F_i and F_j are the respective marginal distributions

Sklar's Representation theorem

- $F(x_1, x_2, \dots, x_n) = C(F_1(x_1), F_2(x_2), \dots, F_n(x_n))$
- $P(X_1 \le x_1, \dots, X_n \le x_n) = C(P(X_1 \le x_1), \dots, P(X_n \le x_n))$
- $C(u, v) = F(F_X^{-1}(u), F_Y^{-1}(v))$

Copula conditions

1) $\lim_{u_k \to 0} C(u_1, u_2) = 0$ for $k = 1, 2$
2) $\lim_{u_1 \to 1} C(u_1, u_2) = u_2$
3) $C(U_1, U_2) - C(u_1, U_2) - C(U_1, u_2) + C(u_1, u_2) \ge 0$ for $u_1 \le U_1$, $u_2 \le U_2$

- non-decreasing, right-continuous.

Invariance Property

- If $\vec{X}$ has copula C, and $T_1, \dots, T_n$ are strictly increasing $(T_1(x_1), \dots, T_n(x_n)) \sim$ copula C, so copula holds under log, inflation, etc.

Frechet Bounds

- Frechet lower bound: $L_F(u_1, \dots, u_n) = \max\left(\sum_{k=1}^{n} u_k - (n-1), 0\right)$
- Frechet upper bound: $U_F(u_1, \dots, u_n) = \min(u_1, \dots, u_n)$

Survival Copula

- $\bar{F}_i(x) = 1 - F_i(x) = S_i(x)$
- $\bar{F}(x_1, x_2) = P(X_1 > x_1, X_2 > x_2) = \bar{C}(\bar{F}_1(x_1), \bar{F}_2(x_2))$
- $\bar{C}(1-u, 1-v) = 1 - u - v + C(u, v)$

Comonotonicity copula

- Frechet Upper bound $- M(u, v) = \min(u, v)$

Countermonotonicity copula

- $W(u, v) = \max(u + v - 1, 0)$

M5

Lower tail

$\lambda_L = \lim_{u \to 0^+} \frac{C(u, u)}{u}$

Upper tail

$\lambda_U = \lim_{u \to 1^-} \frac{\bar{C}(1-u, 1-u)}{1-u} = \lim_{u \to 0^+} \frac{\bar{C}(u, u)}{u}$

Types of Copula

Gaussian (aka Normal) Copula

- $C(u_1, \dots, u_n) = \Phi_\Sigma(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_n))$

Independance Copula

- $F(x_1, x_2, \dots, x_n) = F_1(x_1) \cdot F_2(x_2) \cdots F_n(x_n)$
- $C(u_1, u_2, \dots, u_n) = u_1 \cdot u_2 \cdot \dots \cdot u_n$

Archimedean copula

- $C(u_1, \dots, u_n) = \psi^{-1}(\psi(u_1) + \dots + \psi(u_n))$
- $\psi(1) = 0$ (ψ is a "generator")
- ψ is strictly decreasing
- ψ is convex

Clayton Copula

- $C(u_1, u_2) = (u_1^{-\theta} + u_2^{-\theta} - 1)^{-\frac{1}{\theta}}$
- Archimedean with type
  - $\psi(t) = \frac{1}{\theta}(t^{-\theta} - 1)$
  - $\psi^{-1}(s) = (1 + \theta s)^{-\frac{1}{\theta}}$

Frank Copula

- $C(u_1, u_2) = -\frac{1}{\theta} \log\left(1 + \frac{(e^{-\theta u_1} - 1)(e^{-\theta u_2} - 1)}{e^{-\theta} - 1}\right)$
- Archimedean type
  - $\psi(t) = -\log\left(\frac{e^{-\theta t} - 1}{e^{-\theta} - 1}\right)$
  - $\psi^{-1}(s) = -\frac{1}{\theta} \log(1 + e^{-s}(e^{-\theta} - 1))$

Gumbel (-Hougard) Copula

- $C(u_1, u_2) = \exp(-((-\log u_1)^\theta + (-\log u_2)^\theta)^{1/\theta})$
- $\psi(t) = (-\log(t))^\theta$
- $\psi^{-1}(s) = \exp(-s^{1/\theta})$

Sklar's theorem

- For any multivariate CDF $F(x, y, z)$ with continuous marginal $F_1(x), F_2(y), F_3(z)$ there exists a copula C $F(x, y, z) = C(F_1(x), F_2(y), F_3(z))$
- If we already know the joint dist^n $F(x, y)$ and its marginals $F_X(x)$, $F_Y(y)$ then:

$C(u, v) = F(F_X^{-1}(u), F_Y^{-1}(v))$

$u = F_X(x) \Rightarrow x = F_X^{-1}(u)$

$v = F_Y(y) \Rightarrow y = F_Y^{-1}(v)$

CDFs

$0 < U, V < 1$

Individual Claim Size Modelling

Censoring

- Truncated: observation is NOT recorded at or below/above a certain point.
- Censoring: observation is recorded as c when it is above/below c but c is its exact value when below/above

Before Modelling:

- Summarize the data
- Visualize the data
- Possibly log the data if heavy tail
- Understand data collection procedures
- Investigate unusual features (outliers, leverage points)

Likelihood formulation (left-truncation, right-censoring)

- Observations are represented as

(t_i, x_i, δ_i)

- $\to t_i$ is the left truncation point
- $\to x_j$ is the claim value that produced the data point
- $\to \delta_j$ is the indicator whether the limit has been reached.

Moments

- First Moment: mean
- Second moment: variance
- Third moment: skewness
- Fourth moment: Kurtosis

M3

Distributions

Gamma $\Gamma(\alpha, \beta)$: $g(y) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y}$, $E[y] = \frac{\alpha}{\beta}$

$Var(x) = \frac{\alpha}{\beta^2}$, $M_Y(t) = \left(\frac{\beta}{\beta - t}\right)^\alpha$, skewness $\varsigma = \frac{2}{\sqrt{\alpha}}$

$E[Y^k] = \frac{\Gamma(\alpha + k)}{\Gamma(\alpha)\beta^k}$, $\alpha = 1$: exponential

Inverse Gaussian $IG(\alpha, \beta)$: $g(y) = \frac{\alpha y^{-\frac{3}{2}}}{\sqrt{2\pi\beta}} e^{-\frac{(\alpha - \beta y)^2}{2\beta y}}$

$E[X] = \frac{\alpha}{\beta}$, $Var(x) = \frac{\alpha}{\beta^2}$, $M_Y(t) = e^{\alpha(1 - \sqrt{1 - 2t/\beta})}$, $t \le \frac{\beta}{2}$

$\varsigma = \frac{3}{\sqrt{\alpha}}$

Weibull (τ, c): $g(y) = c\tau (cy)^{\tau - 1} e^{-(cy)^\tau}$

$E[y] = \frac{\Gamma(1 + \frac{1}{\tau})}{c}$, $Var(x) = \frac{\Gamma(1 + \frac{2}{\tau})}{c^2} - \mu_x^2$

no MGF for $\tau < 1$, $t > 0$

Lognormal $LN(\mu, \sigma^2)$: $E[X] = e^{\mu + \frac{\sigma^2}{2}}$, No MGF $t > 0$

$Var(x) = e^{2\mu + \sigma^2}(e^{\sigma^2} - 1)$, $\varsigma = \sqrt{(e^{\sigma^2} + 2)(e^{\sigma^2} - 1)}$

Log-gamma $\log(Y) \sim \Gamma(\gamma, c)$: $E[Y] = \left(\frac{c}{c-1}\right)^\gamma$, $c > 1$

$g(y) = \frac{c^\gamma}{\Gamma(\gamma)} (\log(y))^{\gamma - 1} y^{-(c+1)}$, $Var(x) = \left(\frac{c}{c-2}\right)^\gamma - \mu_x^2$, $c > 2$

No MGF for $t > 0$

Pareto (θ, α): $g(y) = \frac{\alpha}{\theta}\left(\frac{y}{\theta}\right)^{-(\alpha + 1)}$, $y > \theta$, $\alpha > 0$

$E[X] = \frac{\theta\alpha}{\alpha - 1}$ for $\alpha > 1$, $Var(x) = \frac{\alpha\theta^2}{(\alpha - 1)^2(\alpha - 2)}$, $\alpha > 2$

No MGF $t > 0$

Fitting Distributions

Method of Moments

- Choose m moments
- Build a system of m equations, matching empirical moments
- Solve system of equations

MLE

- Likelihood function: $L(\theta; x) = \prod_{i=1}^{n} f(x_i; \theta)$
- $\ell(\theta; x) = \log(L(\theta; x))$
- derive $\frac{\partial \ell(\theta; x)}{\partial \theta} = 0$

Zero-inflated severity model $X = IB$

- $X = IB$, I is an indicator
  $\to P(I = 1) = q$ and $P(I = 0) = 1 - q$
  B is the claim amount given if $I = 1$
- $F_X(x) = 1 - q + q P(B \le x)$
- $M_X(t) = 1 - q + q E[e^{tB}]$
- $E[X] = E[E[X|I]] = E[X|I=1] \cdot P[I=1] = qE(B)$
- $Var(x) = q(1-q)(E[B])^2 + qVar(B)$
- $E[X|I] = I \cdot E[B]$
- $Var(X|I) = I^2 \cdot Var(B)$

Fixed Claim: $B = b$

- $X = Ib$

$X = Ib = \begin{cases} b, & \text{prob } p \\ 0, & \text{prob } 1-p \end{cases}$

- $E[X] = bP$
- $Var(X) = b^2 Var(I) = b^2 p(1-p)$

Akaike Information Criteria

$AIC = -2\ell + 2d$

$\ell = \max$ log Likelihood

$d = \#$ parameters estimated

Bayesian Information Criteria

$BIC = -2\ell + \log(n) d$

$n = \#$ of parameters

Deductible

- insurer pays claims above d

Stop-loss $Y = \begin{cases} 0 & X \le d \\ X - d & X > d \end{cases} = \max(X - d, 0)$

Limit

- insurer pays claims less than M

$Y = \begin{cases} X, & X \le M \\ M, & X > M \end{cases} = \min(X, M)$

Stop loss

- $Y = \min(\max(X - d, 0), M)$

$= \begin{cases} 0, & X \le d \\ X - d, & d < X < d + M \\ M, & X \ge d + M \end{cases}$

Reinsurance

Proportional

- quota share $\to$ proportion is same for all risks
- surplus $\to$ proportion varys from risk to risk
  - Insurer: $Y = \alpha X$
  - Reinsurer: $Z = (1 - \alpha) X$
  - $\mu_Y = \alpha\mu_X$, $\sigma_Y^2 = \alpha^2\sigma_X^2$, $\varsigma_Y = \varsigma_X$

Non-proportional

- excess of loss: on each individual (X_i)
- stop loss: on aggregate loss (S_i)
  - Insurer $Y_i = \min(x_i, d)$ Reinsurer pays excess over
  - Reinsurer $Z_i = (X_i - d)_+$
- $\to$ Insurer $Y_i = \min(x_i, d) + (x_i - d - m)_+$
- $\to$ Reinsurer $Z_i = \min((x_i - d)_+, m)$
- $\to$ Reinsurer limits payments to M

$Y = \sum_{i=1}^{N} Y_i$, $Z = \sum_{i=1}^{N} Z_i$

M4

Panjer's recursive

- Let S have a compound distribution on Y, $\sum_{i}^{N} Y_i$
  - $\to N$ belongs to (a, b) class
  - $\to Y$ are identically distributed, non-negative

$f_S(s) = \frac{1}{1 - ag_0} \sum_{j=1}^{s} \left(a + \frac{bj}{s}\right) g_j f_S(s - j)$, $s = 1, 2, \dots$

$f_S(0) = \begin{cases} P(N=0), & g_0 = 0 \\ M_N(\log(g_0)), & g_0 > 0 \end{cases}$

Translated Gamma and LN

Fitting a translated Gamma (k, γ, c) to a Comp Pois$(\lambda v, G(x))$

- $E[S] = \lambda v E[Y] = k + \frac{\gamma}{c}$
- $Var(S) = \lambda v E[Y^2] = \frac{\gamma}{c^2}$
- $\varsigma_S = \frac{E[Y^3]}{\sqrt{\lambda v E[Y^2]^3}} = 2\gamma^{-\frac{1}{2}}$

Solve for γ, c, k in terms of $E[S]$, $Var(S)$, ς_S.

Approximations for Compound Distributions

Distribution Classes

Panjer distributions

- $(a, b, 0)$ class
  - $\to P(N=k) = \left(a + \frac{b}{k}\right) P(N = k-1)$, $k = 1, 2, 3, \dots$
  - $\frac{p_k}{p_{k-1}} = \left(a + \frac{b}{k}\right)$

Members:	a	b	$P(N=0)$
Poisson (λ)	0	λ	$e^{-\lambda}$
NB (γ, p)	p	$(\gamma - 1)p$	$(1-p)^\gamma$
Binom (v, p)	$\frac{-p}{1-p}$	$\frac{(v+1)p}{1-p}$	$(1-p)^v$

$(a, b, 1)$ class

$\frac{p_k}{p_{k-1}} = a + \frac{b}{k}$ for $k = 2, 3, \dots$

Zero truncated Distributions

- Set $p_0 = 0$ in the $(a, b, 0)$ class

We get $p_k^T = \begin{cases} 0 & k = 0 \\ \frac{p_k}{1 - p_0} & k = 1, 2, \dots \end{cases}$

Zero modified dist^n

- Set $p_0 \equiv p_0^M$ $(0 < p_0^M < 1)$ in $(a, b, 1)$ class

$p_k^M = \begin{cases} p_0^M & k = 0 \\ \frac{1 - p_0^M}{1 - p_0} p_k & k = 1, 2, 3, \dots \end{cases}$

Extreme Value Theory

Measures of tail weight

- log-log plot: linear decrease for heavy tails
- Mean excess function: linear increase for heavy tails
- Existence of moments: fewer finite for heavier tails
- Limiting density ratio
  - $\lim_{x \to \infty} \frac{S_A(x)}{S_B(x)} = \lim_{x \to \infty} \frac{f_A(x)}{f_B(x)}$
- Hazard rate function: $h(x) = \frac{f(x)}{1 - F(x)}$ constant for exponential, decreasing for heavier tails.

M6

n-block maximum

- $X_{1,n} = \min(X_1, \dots, X_n)$
- $X_{n,n} = \max(X_1, \dots, X_n)$

GEV

$H_{\gamma; \mu, \sigma}(x) = \begin{cases} e^{-\left(1 + \gamma\frac{x - \mu}{\sigma}\right)^{-\frac{1}{\gamma}}} & \gamma \ne 0 \\ e^{-e^{-\frac{x - \mu}{\sigma}}} & \gamma = 0 \end{cases}$

case 1: $\gamma < 0$

$\to$ upper bounded Weibull $x < \mu - \frac{\sigma}{\gamma}$ (small tail)

Case 2: $\gamma = 0$

$\to$ Gumbel with $x \in \mathbb{R}$ (medium tail)

Case 3: $\gamma > 0$

$\to$ Gumbel with $x > \mu - \frac{\sigma}{\gamma}$ (heavy tail)

Steps

- Find a_n and b_n: $\lim_{n \to \infty} P\left(\frac{X_{n,n} - b_n}{a_n} \le x\right) = \lim_{n \to \infty} (F(a_n x + b_n)^n)$

range of x: $z \ge \lim_{n \to \infty} \left(\frac{-b_n}{a_n}\right)$, $n(1 - F(a_n x + b_n)) = \tau(x)$

where $H(x) = e^{-\tau(x)}$ $\to$ to find a_n, b_n

Asymptotic properties of the tail

- end of life situation: $u_\varepsilon = F^{-1}(1 - \frac{1}{\varepsilon})$
- the tail is the dist^n past this point 'u_ε' and is characterised with the df of the excess over threshold u (threshold exceedance)
  $F_u(x) = P(Y - u \le x \mid Y > u) = \frac{F(x + u) - F(u)}{1 - F(u)}$
  $0 \le x < x_F - u$
- For high 'u' we get GPD

Generalised Pareto Distribution

- $G_{\gamma, \sigma}(x) = \begin{cases} 1 - (1 + \gamma\frac{x}{\sigma})^{-\frac{1}{\gamma}} & \gamma \ne 0 \\ 1 - e^{-\frac{x}{\sigma}} & \gamma = 0 \end{cases}$

$\gamma < 0$: Pareto type II $x \in (0, \frac{\sigma}{|\gamma|})$ light tail

$\gamma = 0$: exponential $x \in (0, \infty)$

$\gamma > 0$: Pareto $x \in (0, \infty)$ heavy tail

GPD central moment

(in excess of u)

- $E[X] = \frac{\sigma}{1 - \gamma}$, $\gamma < 1$
- $Var(X) = \frac{\sigma^2}{(1 - \gamma)^2(1 - 2\gamma)}$, $\gamma < \frac{1}{2}$

$\frac{d}{d\theta} a^\theta = \log(a) a^\theta$

(1) Weibull $(\gamma < 0)$

- limiting distribution of normalised maximum has $\gamma < 0$ if: it has a finite endpoint $(x_F < \infty$, where

$x_F = \sup\{x \in \mathbb{R} : F(x) < 1\}$

(2) Gumbel $(\gamma = 0)$

Limiting distribution of normalised maximum has $\gamma = 0$ if it has finite moments $E[X^k] < \infty$ for any k

(3) Frechet $(\gamma > 0)$

- $L(x)$ is slowly varying at ∞ if $\lim_{x \to \infty} \frac{L(tx)}{L(x)} = 1$, $t > 0$
- $h(x)$ is regularly varying at ∞ with index ρ if $\lim_{x \to \infty} \frac{h(tx)}{h(x)} = t^\rho \Leftrightarrow h(x) = x^\rho L(x)$, $t > 0$

$E[X^k] = \infty$ for $k > \frac{1}{\gamma}$

Choice of u

- We need to eliminate the tail $\bar{F}(x + u) = \bar{F}(x) \cdot \bar{F}(u)$ for large u, and all $x > 0$
- Prob of exceeding u: $\widehat{F(u)} = \frac{n_u}{n}$
  then for a given u: $\bar{F}(u) = G_{\hat{\gamma}, \hat{\sigma}}(x)$
  - $\hat{\gamma}(u)$ and $\hat{\sigma}(u)$ are estimated in the data in the tail $> u$
- Large u leads to better approximation from a distributional perspective but it reduces the amount of data to estimate $\bar{F}(u)$, γ, σ

(1) Frechet $\gamma > 0$

- The limiting distribution of the normalised maximum from an underlying df $(F_X(x))$ is of Frechet type $(\gamma > 0)$ iff $\bar{F}(x) = 1 - F(x) = x^{-\frac{1}{\gamma}} L(x)$